PYL1002: Waves and Oscillations
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Tutorial - 7
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Waves at Boundaries and Impedance Matching

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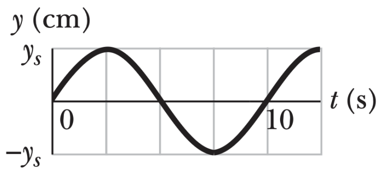
1 *Characterize this Traveling Wave [Quick]

A sinusoidal transverse wave of wavelength 20 cm travels along a string in the positive x direction. The displacement y of the string particle at $x = 0$ as a function of time is given in the figure below.



The scale of the vertical axis is set by $y_s = 4$ cm.

a Write down the wave solution $y(x, t)$. In particular, identify the standard parameters of the wave, such as its angular frequency ω, wavenumber k, speed v, amplitude A, and initial phase ϕ.

b At $t = 0$, sketch a plot of y versus x.

c What is the transverse velocity of the particle at $x = 0$ when $t = 5$ s?

2 *Solving the Wave Boundary Conditions [Foundational]

In class, we analyzed waves at the boundary of two different media. We looked at two strings, one with tension T_1 and mass density μ_1 to the left of the junction at $x = 0$ knotted with another on the right side with tension T_2 and mass density μ_2. The traveling wave solutions we looked as had the following form split into incident (i), reflecting (r), and transmitted (t) waves.

$$y(x,t) = \begin{cases} y_L(x,t) = y_i(x,t) + y_r(x,t), & x \leq 0 \\ y_R(x,t) = y_t(x,t), & x \geq 0 \end{cases}$$

We imposed two boundary conditions to have a physically reasonable model at the junction:

1. **Continuity:** The wave motion from the left must match from the right so the string does not break,

$$y_L(0,t) = y_R(0,t)\ . \tag{1}$$

2. **Force Balance:** The junction is a microscopic massless point and hence the net forces on it must balance to ensure a finite acceleration. For small angles, this condition reads the following,

$$T_1 \frac{\partial y_L}{\partial x}\bigg|_{x=0} = T_2 \frac{\partial y_R}{\partial x}\bigg|_{x=0} . \tag{2}$$

Using traveling wave solution forms for the incident (i), reflecting (r), and transmitted (t) waves, such as,

$$y_i(x,t) = f_i(t - \frac{x}{v_1})\ , \text{and similar for } y_r \text{ and } y_t\ . \tag{3}$$

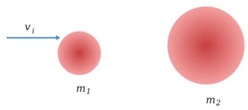
Solve the boundary condition equations to derive the "standard result" on reflection and transmission that we worked with in class,

$$y_r = \left(\frac{Z_1 - Z_2}{Z_1 + Z_2}\right) y_i\ , \quad \text{and} \quad y_t = \left(\frac{2Z_1}{Z_1 + Z_2}\right) y_i \tag{4}$$

where $Z = T/v$ is the *impedance* of a given medium. *Hint:* The force balance condition involves derivatives of the wave shape functions, a quick integration might help.

3 *Impedance Matching in Elastic Collisions [Substantive]

In class, we discussed how mechanical elastic collisions closely resemble the physics of waves at a boundary. Let us continue our exploration of this idea. Consider the following elastic collision where the incoming ball as a speed v_i and the second ball starts from rest.



a Use principles of conservation of energy conservation and momentum conservation for the elastic collision to show that the speed of the two balls after collision satisfy,

$$v_r = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) v_i \,, \quad \text{and} \quad v_t = \left(\frac{2m_1}{m_1 + m_2}\right) v_i \tag{5}$$

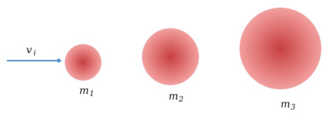
where v_r is the velocity of the first ball after collision and v_t is the post-collision velocity of the second ball.

b Using your results above, determine the fraction of the initial kinetic energy of the first ball that is transferred to the second ball after the collision. Express your answer as $T = E_t/E_i$ which is called as the "transfer ratio" and show that

$$T = \frac{4m_1 m_2}{(m_1 + m_2)^2} \,. \tag{6}$$

Sketch or qualitatively describe how the transmission fraction T behaves when $m_2 \gg m_1$ and when $m_2 \ll m_1$. When is energy transfer maximized?

c Now suppose we wish to transfer energy from a ball of mass m_1 to a ball of mass m_3, but we insert an intermediate ball of mass m_2 between them. Assume the balls collide sequentially as shown below.



Using the transmission result from the previous part, write an expression for the total fraction of the initial kinetic energy of the first ball that is ultimately transferred to the third ball after the two collisions.

d Treating m_1 and m_3 as fixed, determine the value of the intermediate mass m_2 that maximizes the energy transferred to the third ball. Show that the optimal choice for m_2 is the geometric mean of the edge masses,

$$m_2 = \sqrt{m_1 m_3} \,. \tag{7}$$

This result is closely related to the principle of *impedance matching* in wave physics as we have been discussing. *Hint:* From part (b), you found the fraction of kinetic energy transferred from one ball to the next in a single elastic collision. In the three–ball setup, energy must first be transferred from m_1 to m_2, and then from m_2 to m_3. Use the result from part (b) to write the fraction of the initial energy that reaches m_3 after the two successive collisions. How are the two transmission factors related? Taking logs may prove extremely helpful.

e Finally, imagine inserting several intermediate balls between m_1 and m_N so that energy is transferred through a cascade of elastic collisions. Suppose the masses form a sequence $m_1, m_2, m_3, \ldots, m_N$.

Argue/show that if we wish to maximize the transfer energy at each collision, the ratio between successive masses should remain approximately constant. Show that the masses should therefore follow a geometric progression of the form $(i = 1, 2, \cdots, N)$,

$$m_n = m_1 \left(\frac{m_N}{m_1}\right)^{\frac{n-1}{N-1}} \,. \tag{8}$$

You may neglect multiple collisions occuring from any rebounding balls.

4 *Megaphones are Conical [Experiment]

Megaphones help project sound over large distances because their conical shape gradually matches the acoustic impedance between the sound source and the surrounding air, allowing sound waves to propagate more efficiently.

Using a sheet of thicker card paper, construct two simple sound guides of roughly the same length:

(i) a **cylindrical tube**

(ii) a **conical tube**

Use one smartphone to generate a steady tone by using the **Phyphox** → **Tone Generator** (for example $f = 400\,\text{Hz}$) at a nominal volume. Place the phone speaker at the narrow entrance of the tube. Use a second smartphone running **Phyphox** → **Audio Amplitude** to measure the sound level (in dB) at the output of the tube. You may see `http://hyperphysics.phy-astr.gsu.edu/hbase/Sound/db.html` for a quick intro to the decibel sound scale and what it means.

Measure the detected sound level for the following cases:

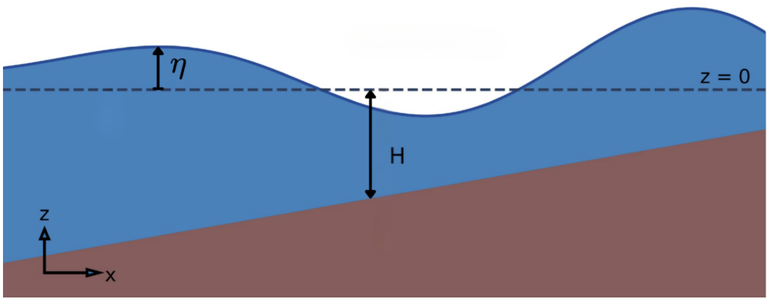
Setup	Measured Sound Level (dB)
Cylindrical tube	
Conical tube	

Which configuration produces the highest output amplitude? Briefly explain how the conical shape may help transmit sound more efficiently in terms of impedance matching and reduced reflections.

5 Tsunami Shoaling: Waves in Shallow Water [Substantive]

In the previous tutorial, we explored how water waves travel in the ocean under the assumption of *deep water*, that is, $H >> \lambda$ where H is the depth of the ocean floor and λ is the wavelength of the wave. We found a peculiar dispersion relation $\omega = \sqrt{gk}$ for such deep water ocean waves, and found that their amplitude fell exponentially with depth inside the ocean.

Today, we will look at **Tsunami Waves**. As ocean tsunami waves near the coastline, something formidable happens. Entering shallow coastal waters, the speed of the tsunami drops, while causing its height to increase dramatically – sometimes by several meters – turning it into a devastating force. This is known as *Tsunami Shoaling* and such waves are called **Shallow Water Waves**. We will investigate how having the opposite condition $H << \lambda$ changes how the dispersion relation looks in shallow water and how it implies tsunamis to be so tall.



As before, we consider waves traveling along the x direction on the surface of water. The vertical coordinate z measures height, with $z = 0$ representing the undisturbed surface of the water and $z = -H$ representing the ocean/coastal bed. As you can imagine, the height H decreases as you approach the shore, so $H \equiv H(x)$. The vertical displacement of the surface is denoted $\eta(x, t)$. The water velocity is drivable from a scalar velocity potential $\phi(x, z, t)$, which for an incompressible fluid satisfies the "wave" equation,

$$\vec{v} = \underbrace{\left(\frac{\partial \phi}{\partial x}\right)}_{v_x} \hat{i} + \underbrace{\left(\frac{\partial \phi}{\partial z}\right)}_{v_z} \hat{k} \quad \underbrace{\Longrightarrow}_{\text{incompressibility}} \quad \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$$

a Assume a wave-like form for the velocity potential $\phi(x, z, t) = A(z)\cos(kx - \omega t)$ and show that under our wave equation, $A(z)$ satisfies

$$\frac{d^2 A(z)}{dz^2} - k^2 A(z) = 0.$$

b In the deep-water problem we required the motion to vanish far below the surface. In the present case the ocean has a bottom boundary at $z = -H$. Physically, water cannot flow through the solid ocean floor which acts as a hard boundary. What condition must the vertical velocity component v_z satisfy at the ocean floor? Use this condition to show[1] that $A(z) = C \cosh[k(z+H)]$ for some constant C.

c Same as last week again: Let the surface displacement be $\eta(x,t) = B\sin(kx - \omega t)$. The vertical motion of the surface must match the vertical velocity of the fluid at the surface:

$$\frac{\partial \eta}{\partial t} = \left.\frac{\partial \phi}{\partial z}\right|_{z=0}.$$

Enforce this surface matching condition above to determine the relation between B and A_0, where $A_0 = A(0)$.

d The pressure at the ocean surface must equal atmospheric pressure. For small waves this leads to the linear Bernoulli condition,

$$\frac{\partial \phi}{\partial t} + g\eta = 0 \qquad \text{at } z = 0.$$

Use this condition together with your earlier results to show that shallow-water waves satisfy the dispersion relation $\omega^2 = gk\tanh(kH)$. See footnote to know more about hyperbolic trigonometry.

e We now consider the *shallow water* limit, where the wavelength of the wave is much larger than the ocean depth $kH << 1$. Using your expression for $A(z)$, argue that in this limit the horizontal velocity v_x of the fluid is approximately constant to leading order. Physically interpret this result.

f Show that in the shallow water limit $\tanh(kH) \approx kH$ to leading order which gives the dispersion relation $\omega^2 \approx gHk^2$.

g The speed of a wave is given by $v = \omega/k$. Show that shallow-water waves travel with speed $v = \sqrt{gH}$ independent of its wavelength and determined only by gravity. Compare this with the deep-water result from last class. Does this explain how tsunamis slow down close to the coast?

A wave also carries energy. For small surface waves the energy per unit horizontal area scales approximately as $E \propto \rho g B^2$, where B is the wave amplitude on the surface and ρ is the density of water. The energy transported by the wave per unit time across a vertical slice of the ocean is the *energy flux*, Flux $= Ev$

h Suppose the ocean depth $H(x)$ changes slowly as the wave approaches the coast. In this case very little of the wave is reflected. Explain briefly why a gradual change of the ocean floor acts like an impedance-matched boundary, allowing the wave energy to continue propagating forward.

i If little energy is reflected, the energy flux carried by the wave must remain approximately constant as it propagates toward shore, Flux $= Ev \approx$ constant. Using the expressions above, show that

$$B^2 v = \text{constant}.$$

j For shallow water waves, show that the amplitude of the wave must scale approximately as

$$B \propto H^{-1/4},$$

and qualitatively explains why this effect causes tsunami waves to grow dramatically in height as they approach shallow coastal regions.

[1] You may recall from your time in the midsem exam that the hyperbolic cosine is defined as $\cosh x = (e^x + e^{-x})/2$. Similarly, hyperbolic sine is $\sinh x = (e^x - e^{-x})/2$, and hyperbolic tan is $\tanh x = \sinh x / \cosh x$.